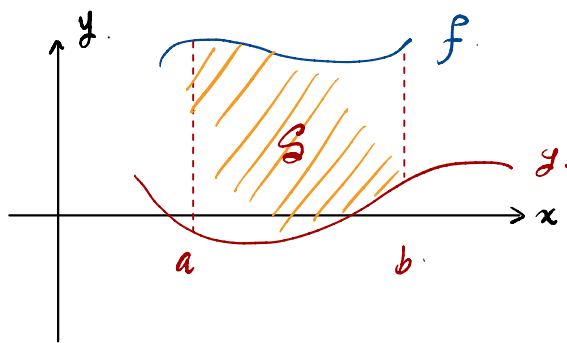
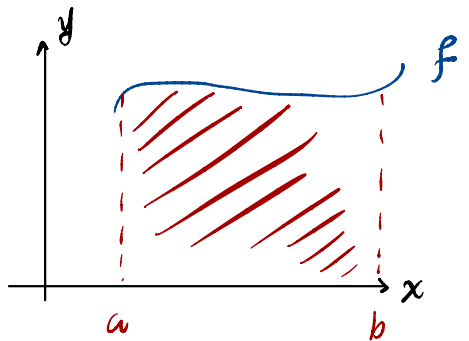
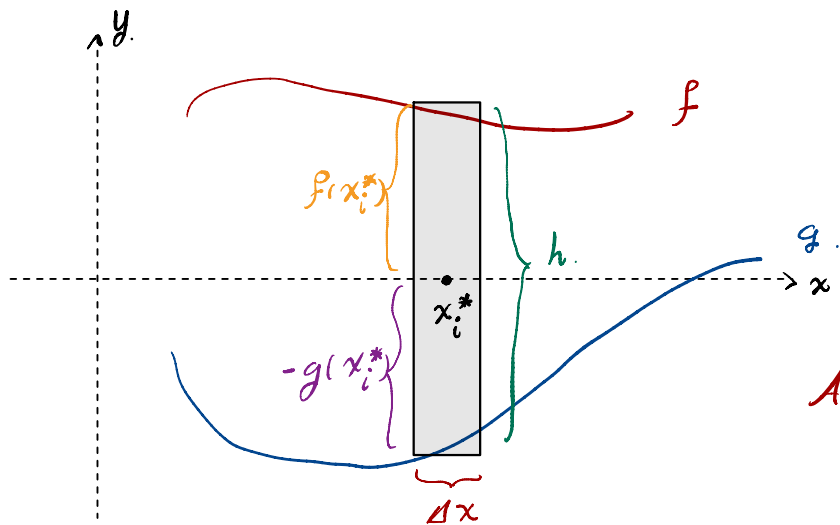


5.1. Areas between Curves.



Consider the region S that lies between two curves $y=f(x)$ and $y=g(x)$ and the vertical lines $x=a$ and $x=b$. where f and g are continuous functions and $f(x) \geq g(x)$.



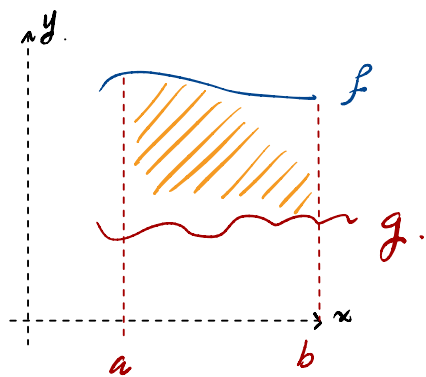
$$A_R = \Delta x \cdot h = \Delta x \{ f(x_i^*) - g(x_i^*) \}.$$

The approximation appears to become better and better as $n \rightarrow \infty$ (where n is the number of rectangles). Therefore, we define the area A of the region S as

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n [f(x_i^*) - g(x_i^*)] \Delta x.$$

Def. The area A of the region bounded by the curves $y=f(x)$ and $y=g(x)$ and lines $x=a$, $x=b$ where f, g are continuous and $f(x) \geq g(x)$ for all $x \in [a, b]$ is

$$A = \int_a^b [f(x) - g(x)] dx.$$

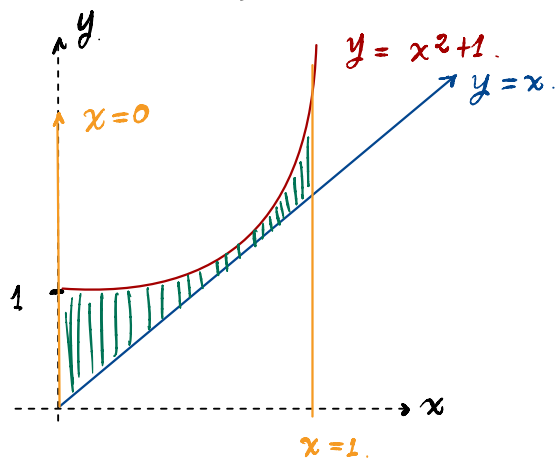


In the case where $f, g \geq 0$ then

$$\begin{aligned} A &= [\text{area under } y=f(x)] - [\text{area under } y=g(x)] \\ &= \int_a^b f(x) dx - \int_a^b g(x) dx \\ &= \int_a^b [f(x) - g(x)] dx. \end{aligned}$$

Example. Find the area of the region bounded above by $y=x^2+1$, bounded below $y=x$, and bounded on the sides by $x=0$ and $x=1$.

Sln:



limits are $0 \leq x \leq 1$.

$$\begin{aligned} A &= \int_0^1 [(x^2+1) - (x)] dx \\ &= \int_0^1 (x^2 - x + 1) dx \\ &= \left. \frac{x^3}{3} - \frac{x^2}{2} + x \right|_0^1 = \frac{1}{3} - \frac{1}{2} + 1 = \frac{5}{6}. \end{aligned}$$

Example. Find the area of the region enclosed by the parabolas

$$y=x^2 \text{ and } y=2x-x^2.$$

Sln: We first find the points of intersection of the parabolas by solving their equation simultaneously.

$$x^2 = 2x - x^2.$$

$$2x^2 - 2x = 0$$

$$x(x-1) = 0 \Rightarrow x=0, x=1.$$

$$\text{Let } x = \frac{1}{2} \quad f(x) = x^2 \Rightarrow f\left(\frac{1}{2}\right) = \frac{1}{4}$$

$$g(x) = 2x - x^2 \Rightarrow g\left(\frac{1}{2}\right) = 1 - \frac{1}{4} = \frac{3}{4}.$$

Since $g\left(\frac{1}{2}\right) \geq f\left(\frac{1}{2}\right)$ then $g(x) \geq f(x)$ for all $x \in [0, 1]$.

$$A = \int_0^1 \{g(x) - f(x)\} dx = \int_0^1 \{(2x - x^2) - (x^2)\} dx = \int_0^1 (2x - 2x^2) dx$$

$$= x^2 - \frac{2}{3}x^3 \Big|_0^1 = 1 - \frac{2}{3} = \frac{1}{3}.$$

Example. Find the approximate area of the region bounded by the curves $y = \frac{x}{\sqrt{x^2+1}}$ and $y = x^4 - x$.

Sln: We can't solve the equation $x^4 - x = \frac{x}{\sqrt{x^2+1}}$ we can approximate the integration limits using geoalgebra.

$$A = \int_0^{1.18} \left\{ \frac{x}{\sqrt{x^2+1}} - (x^4 - x) \right\} dx = \int_0^{1.18} \frac{x}{\sqrt{x^2+1}} dx - \int_0^{1.18} (x^4 - x) dx.$$

$\int \frac{x}{\sqrt{x^2+1}} dx$ let $u = x^2 + 1$ $du = 2x dx \Rightarrow x dx = \frac{du}{2}$
by the substitution rule

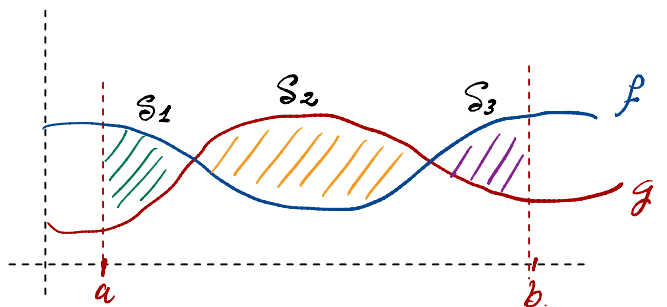
$$\int \frac{x}{\sqrt{x^2+1}} dx = \frac{1}{2} \int \frac{1}{\sqrt{u}} du = \frac{1}{2} \int u^{-1/2} du = \sqrt{u} = \sqrt{x^2+1}.$$

then,

$$\int_0^{1.18} \frac{x}{\sqrt{x^2+1}} dx = \sqrt{x^2+1} \Big|_0^{1.18} = \sqrt{(1.18)^2+1} - 1.$$

$$A = \sqrt{(1.18)^2+1} - 1 - \left(\frac{x^5}{5} - \frac{x^2}{2} \Big|_0^{1.18} \right) = \sqrt{2.39} - 1 - \left(\frac{(1.18)^5}{5} - \frac{(1.18)^2}{2} \right)$$

If we are asked to find the area between $y=f(x)$ and $y=g(x)$ where $f(x) \geq g(x)$ for some values of x but $g(x) \geq f(x)$ for other values of x .



then we split the region S into several regions $S_1, S_2, S_3, \dots$ with areas $A_1, A_2, A_3, \dots$

We then define the area of the region S to be the sum of the areas of the smaller regions. $A = A_1 + A_2 + A_3$.

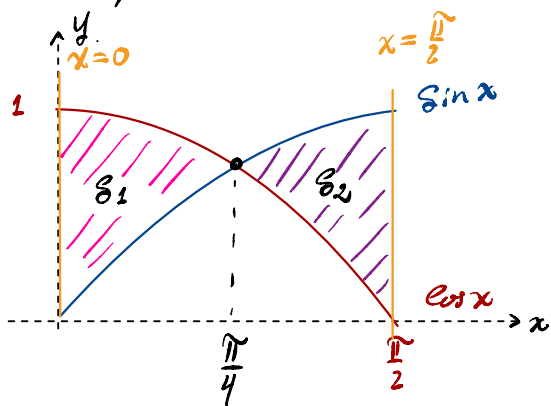
$$|f(x) - g(x)| = \begin{cases} f(x) - g(x) & \text{if } f(x) \geq g(x) \\ g(x) - f(x) & \text{if } g(x) \geq f(x). \end{cases}$$

Def. The area between the curves $y=f(x)$ and $y=g(x)$ and $x=a$, $x=b$ is

$$A = \int_a^b |f(x) - g(x)| dx.$$

Example. Find the area of the region bounded by the curves $y = \sin x$, and $y = \cos x$, $x=0$, $x = \frac{\pi}{2}$.

Sln: The point of intersection occurs when $x = \frac{\pi}{4}$.

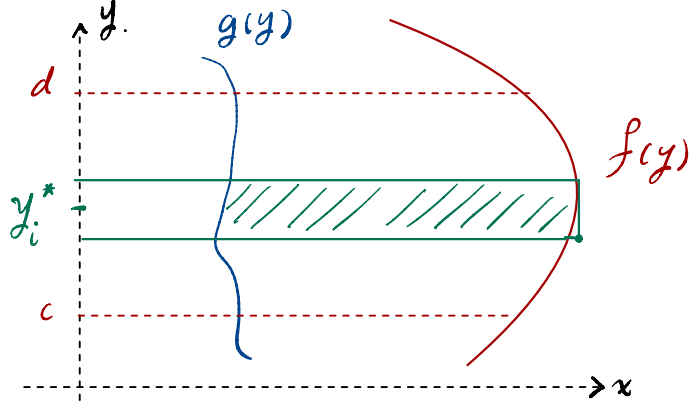


$$\begin{aligned} A &= \int_0^{\frac{\pi}{2}} |\cos x - \sin x| dx = A_1 + A_2 \\ &= \underbrace{\int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx}_{A_1} + \underbrace{\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x - \cos x) dx}_{A_2} \\ &= (\sin x + \cos x) \Big|_0^{\frac{\pi}{4}} + (-\cos x - \sin x) \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\ &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - (0 + 1) \right) + \left(-0 - 1 - \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \right) \right) \\ &= 2\sqrt{2} - 2. \end{aligned}$$

Area between curves: (Integrating with respect y).

Some regions are best treated by regarding x as a function of y .

If the region bounded by the curves $x=f(y)$, $x=g(y)$, $y=c$ and $y=d$ where f, g are continuous functions and $f(y) \geq g(y)$ for $y \in [c, d]$ then



$$A = \int_c^d [f(y) - g(y)] dy.$$

Example. Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

Sln. By solving $\begin{cases} x = y + 1 = f(y) \\ x = \frac{y^2 - 6}{2} = g(y). \end{cases}$

$$y + 1 = \frac{y^2 - 6}{2} \quad (\Rightarrow) \quad 2y + 2 = y^2 - 6$$

$$(\Rightarrow) \quad y^2 - 2y - 8 = 0$$

$$(\Rightarrow) \quad (y - 4)(y + 2) = 0$$

$$(\Rightarrow) \quad y = 4, \quad y = -2$$

$$\text{If } y = 0 \quad \text{then} \quad f(0) = 1$$

$$g(0) = -3$$

then $f(y) \geq g(y)$ for $-2 \leq y \leq 4$.

$$A = \int_{-2}^4 [f(y) - g(y)] dy = \int_{-2}^4 \left[y + 1 - \left(\frac{y^2}{2} - 3 \right) \right] dy$$

$$= \int_{-2}^4 \left(-\frac{y^2}{2} + y + 4 \right) dy$$

$$= -\frac{1}{6} y^3 + \frac{y^2}{2} + 4y \Big|_{-2}^4$$

$$= -\frac{1}{6} (4^3) + \frac{4^2}{2} + 4^2 + \frac{1}{6} (2)^3 - 2 - 8 = 18.$$

Problem for you: Find the area of the region enclosed by $x + 2y = 3$

$y = x$ and $y = \frac{1}{4}x$ using.

a). x as variable of integration

b). y as variable of integration.